

SIT 210

Embedded Systems Development

Task 3.1P

Summary

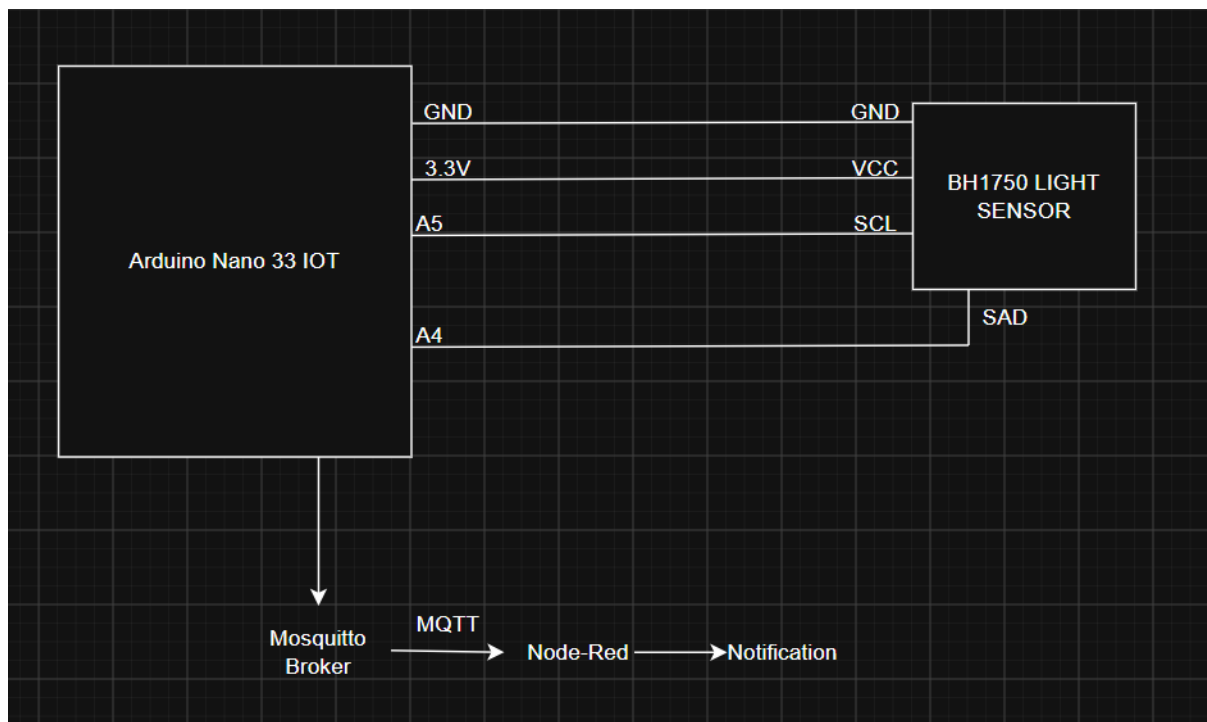
In this task, I created a system to monitor the light level of a terrarium using an Arduino Nano 33 IoT and BH1750 light sensor. The sensor was connected using the I2C pins, including VCC, GND, SDA and SCL. The Arduino reads the light intensity in lux and checks if it is above or below the selected threshold of 50 lux.

When the light level goes above 50 lux, the Arduino sends a `SUNLIGHT_START` message. When the light level goes below 50 lux, it sends a `SUNLIGHT_STOP` message. These messages are published to the MQTT topic `terrarium/light`.

For communication, I installed Mosquitto on my Windows laptop and used it as the MQTT broker. I also found my laptop IPv4 address and used it in the Arduino code so the Arduino could connect to the broker through Wi-Fi. One difficulty was understanding how MQTT works and connecting the Arduino to the correct broker IP and port. We also had to make sure Mosquitto was running and allow port 1883 through the Windows Firewall.

After uploading the code, I tested the sensor by covering it and exposing it to light. The Serial Monitor showed different lux values and successfully displayed `SUNLIGHT_STOP` sent! when the light was low and `SUNLIGHT_START` sent! when the light increased. After this, the next step was connecting Mosquitto to Node-RED using an MQTT input node and checking whether the messages were received in the Debug panel.

Block Diagram



Code for showing readings and connectivity

```
// Wi-Fi details
const char* ssid = "Redmi Note 13 5G";
const char* pass = "akash@123";

// Your laptop IPv4 address
const char* mqtt_server = "10.18.17.121";

const float LIGHT_THRESHOLD = 50.0;
```

```
if (lightMeter.begin()) {
  Serial.println("BH1750 sensor connected!");
} else {
  Serial.println("BH1750 sensor NOT found!");
}
```

```
// Connect to MQTT
if (!mqttClient.connected()) {

    Serial.println("Connecting to MQTT...");

    if (mqttClient.connect("ArduinoTerrariumClient")) {
        Serial.println("Connected to MQTT!");
    } else {
        Serial.print("MQTT connection failed. State: ");
        Serial.println(mqttClient.state());

        delay(5000);
        return;
    }
}
```

Trigger Mechanism

```
// Detect sunlight starting
if (lux > LIGHT_THRESHOLD && !sunlightOn) {

    mqttClient.publish("terrarium/light", "SUNLIGHT_START");

    Serial.println("SUNLIGHT_START sent!");

    sunlightOn = true;
}
```

```
// Detect sunlight stopping
else if (lux <= LIGHT_THRESHOLD && sunlightOn) {

    mqttClient.publish("terrarium/light", "SUNLIGHT_STOP");

    Serial.println("SUNLIGHT_STOP sent!");

    sunlightOn = false;
}
```

Arduino IDE readings

[illegible]

Node-red Readings

DEBUG MESSAGES

all nodes

all

25/8/2026, 10:34:37 pm

node: debug 1

terrarium/light : msg.payload : string[14]

"SUNLIGHT_START"

25/8/2026, 10:34:47 pm

node: debug 1

terrarium/light : msg.payload : string[13]

"SUNLIGHT_STOP"

25/8/2026, 10:34:49 pm

node: debug 1

terrarium/light : msg.payload : string[14]

"SUNLIGHT_START"

25/8/2026, 10:34:59 pm

node: debug 1

terrarium/light : msg.payload : string[13]

"SUNLIGHT_STOP"

25/8/2026, 10:35:01 pm

node: debug 1

terrarium/light : msg.payload : string[14]

"SUNLIGHT_START"

Node-RED

Flow 1

+

-

Deploy

EXPLORER

Search flows

Flows

Flow 1

Subflows

Global Configuration Nodes

PALETTE

filter nodes

common

inject

debug

complete

catch

status

link in

link call

link out

comment

function

function

switch

change

terrarium/light

connected

debug 1

Flow 1

Flow

"30debeb465c5196e"

DEBUG MESSAGES

all nodes

all

25/8/2026, 10:34:37 pm

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terrarium/light : msg.payload : string[14]

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25/8/2026, 10:34:47 pm

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25/8/2026, 10:35:01 pm

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terrarium/light : msg.payload : string[14]

"SUNLIGHT_START"

Physical Circuit

